

Federated Deep Learning for Privacy-Preserving ECG Classification Using an Iterative Error Feedback CNN-Transformer with Token Pruning

Abhinay Kumar*, Sagili Karthik Reddy*, Rapolu Vijay Srikar*, Rongali Sai Krishna Rohith*, Vasundhara†

* Department of ECE

National Institute of Technology

Warangal, India

{ak22ecb0b08, sk22ecb0b02, rv22ecb0b25, rs21ecb0b30}@student.nitw.ac.in

† Department of ECE

National Institute of Technology

Warangal, India

vasundhara@nitw.ac.in

Abstract—Timely and precise classification of ECG signals is essential in detecting critical heart diseases, but centralized storage and processing of ECGs from numerous patients pose significant challenges in terms of privacy concerns, regulatory compliance, and technical complexity. This paper proposes a federated learning approach toward ECG classification where no patient data is shared across institutions. Central to our proposal is a new method called Iterative Error Feedback CNN-Transformer with Token Pruning (IEFCT-TP), where convolutional network learns features locally and transformer analyzes global sequences, uses an attentionscore-guided token pruning technique to remove redundant token representations, and utilizes an iterative error feedback mechanism to integrate prediction residuals with intermediate feature maps. IEFCT-TP performs exceedingly well in both IID and non-IID partitioning schemes in both MIT-BIH Arrhythmia and St. Petersburg INCART databases, obtaining 99.3% accuracy and 98.95% F1-score using the traditional FedAvg aggregation protocol; IEFCT-TP outperforms the CNN, ResNet-1D, LSTM, and Transformer baselines by considerable margins. In addition, our framework shows excellent robustness to data heterogeneity and can accommodate adaptive aggregation rules such as FedAdam, FedAdagrad, and Mean Regularized Multi-Task Learning (MR-MTL). **Index Terms**—Federated Learning, ECG Classification, CNN-Transformer, Token Pruning, Iterative Error Feedback, Privacy Protection, Data Heterogeneity, FedAvg, MR-MTL

Index Terms—Federated Learning, ECG Classification, CNN-Transformer, Token Pruning, Iterative Error Feedback, Privacy Preservation, Non-IID Data, FedAvg, MR-MTL, Arrhythmia Detection

I. INTRODUCTION

Cardiovascular disease (CVD) persists as the primary source of untimely deaths around the world, causing nearly 18 million deaths each year [1]. The electrocardiogram, which continuously records the sequences of electrical impulses that control myocardial activation and deactivation, is the primary medical test used to diagnose arrhythmias, myocardial infarctions, bundle branch blocks, and various conduction abnormalities. Interpretation by human cardiologists is slow, vari-

able among readers, and inherently inadequate to meet the demands of continuously recording wearable technology associated with telemedicine. Deep-learning classifiers offer a more promising alternative, with convolutional neural architectures analyzing beat-level morphology, recurrent and attention networks analyzing rhythm-level temporal structures, and entire models approaching or surpassing the diagnostic accuracy of board-certified cardiologists on selected benchmarks [2].

However, the need for a sufficiently extensive data set to train such classifiers runs counter to laws protecting the medical privacy of patients. For example, data sets large enough to train a generalizable arrhythmia classifier are distributed among different hospitals, research centers, and wearable technology systems. Combining all such records within a single repository violates both the GDPR and HIPAA and their national counterparts regarding restrictions on the transference of biometric data from institution to institution [3].

Federated Learning (FL), introduced by McMahan *et al.* [4], resolves this tension architecturally. Participating institutions—hospitals, clinics, or wearable gateways—retain their data locally and contribute only encrypted model parameter updates to a coordinating server. The server aggregates these updates, broadcasts an improved global model, and repeats the cycle. Raw ECG signals are never transmitted. The scheme preserves data locality while enabling the kind of multi-site, population-diverse training that improves generalization and reduces demographic bias.

Although several recent works have explored federated ECG classification [5]–[8], critical gaps remain: (i) most architectures rely on convolutional networks that cannot efficiently capture long-range inter-beat rhythmic structure; (ii) token-level redundancy in transformer models inflates communication overhead in FL, where bandwidth is a first-class resource; (iii) gradient-level correction mechanisms that exploit prediction residuals to improve convergence under non-IID

distributions have not been explored for federated cardiac AI.

This work addresses all three gaps through the following contributions:

- This paper proposes the *Iterative Error Feedback CNN-Transformer with Token Pruning* (IEFCT-TP), a hybrid architecture that combines multi-scale CNN feature extraction, learnable-positional-encoding transformer encoders, and a principled token pruning mechanism that discards low-salience representations before the attention layers.
- The proposed methodology introduces an *iterative error feedback* (IEF) mechanism in which per-round prediction residuals are projected back into the intermediate feature space, creating a self-corrective learning signal that accelerates convergence and mitigates client-drift under heterogeneous data partitions.
- The proposed framework conducts a comprehensive evaluation across four federated aggregation strategies (FedAvg, FedAdam, FedAdagrad, MR-MTL), two benchmark databases, and both IID and non-IID splits, providing a systematic ablation of the design choices.
- IEFCT-TP achieves state-of-the-art performance of 99.3% accuracy and 98.95% F1-score, outperforming all baselines and recent literature in balanced four-metric evaluation.

The remainder of the paper is organized as follows. Section II reviews related work. Section III describes the federated learning system model. Section IV details the IEFCT-TP architecture and its mathematical formulation. Section V covers the experimental setup. Section VI presents and discusses empirical results. Section VII concludes, and Section VIII outlines future directions.

II. RELATED WORK

A. Deep Learning for ECG Classification

Early automated ECG analysis relied on hand-engineered signal features—wavelet coefficients, RR-interval statistics, spectral band energies—fed into classical classifiers such as SVMs and random forests [18], [19]. While interpretable, these pipelines require domain expertise and degrade noticeably when applied to out-of-distribution recordings. Deep learning architectures learned discriminative representations end-to-end: one-dimensional CNNs extract local morphological features [7], ResNet variants introduced skip-connected residual blocks to enable training of substantially deeper networks [8], [16], and bidirectional LSTMs capture bidirectional temporal dependencies in beat sequences. More recently, pure transformer architectures [13], [14] and hybrid CNN-Transformer designs [16], [17] have demonstrated that multi-head self-attention can model inter-beat relationships that recurrent models struggle to represent at long lags. Sharan *et al.* [15] further showed that processing each ECG lead independently before fusing via attention yields strong multi-label performance on large 12-lead collections.

B. Federated Learning for Cardiac Monitoring

Weimann and Conrad [6] conducted a large-scale study training deep ResNet networks across heterogeneous PhysioNet ECG collections under various FedOpt strategies, finding that group normalization and local fine-tuning after federation are critical for recovering centralized-training performance. Çelik and Güllü [5] provided a direct comparison of FedAvg, FedAdam, FedAdagrad, and FedYogi on the MIT-BIH database, confirming that IID federation closely matches centralized accuracy ($\sim 98.5\%$) while non-IID partitions degrade performance to around 87–90%. Mehta and Rathour [7] demonstrated privacy-preserving ECG classification with minimal accuracy loss in IID settings. Mane *et al.* [8] reported high accuracy using federated ResNet across a five-client cross-institution simulation. ElBedoui [9] uniquely explored the complementary problem of federated *unlearning*, proposing iterative mechanisms to revoke the influence of specific clients’ data post-training. Asif *et al.* [10] proposed weighted aggregation where client contributions are scaled by their relevance to the global model, improving non-IID robustness.

C. Research Gaps Addressed

Despite this progress, three limitations persist. First, token-level redundancy in transformer-based FL has not been addressed: all ECG time-steps are transmitted and processed equally, wasting bandwidth and computation on uninformative segments. Second, the feedback from prediction residuals back into the feature extraction pipeline—a correction mechanism well-studied in signal processing but underexplored in deep federated learning—has not been applied to ECG tasks. Third, no prior work simultaneously evaluates multiple deep architectures, multiple aggregation strategies, IID/non-IID data conditions, and multi-database generalization within a single study.

III. SYSTEM MODEL: FEDERATED LEARNING FRAMEWORK

A. Architecture Overview

The federated system, illustrated conceptually in Fig. III-A, comprises one central parameter server $\mathcal{S}$ and N distributed clients $\{\mathcal{C}_1, \mathcal{C}_2, \dots, \mathcal{C}_N\}$. Each client $\mathcal{C}_i$ holds a private local dataset $\mathcal{D}_i = \{(\mathbf{x}_j, y_j)\}_{j=1}^{n_i}$ of segmented ECG beats with integer class labels $y_j \in \{0, 1, 2, 3, 4\}$ corresponding to the AAMI five-class mapping (N, S, V, F, Q). The server never observes $\mathcal{D}_i$; it only receives model weight tensors transmitted over a secure channel.

B. Federated Training Protocol

A single training *round* t proceeds through the following steps:

(1) **Broadcast.** The server transmits the current global weights $\mathbf{w}^{(t)}$ to all active clients.

(2) **Local optimization.** Each client $\mathcal{C}_i$ initializes its local model from $\mathbf{w}^{(t)}$ and minimizes the local empirical risk over E epochs using stochastic gradient descent:

$$\mathbf{w}_i^{(t+1)} = \mathbf{w}^{(t)} - \eta \nabla \mathcal{L}_i(\mathbf{w}^{(t)}; \mathcal{D}_i), \quad (1)$$

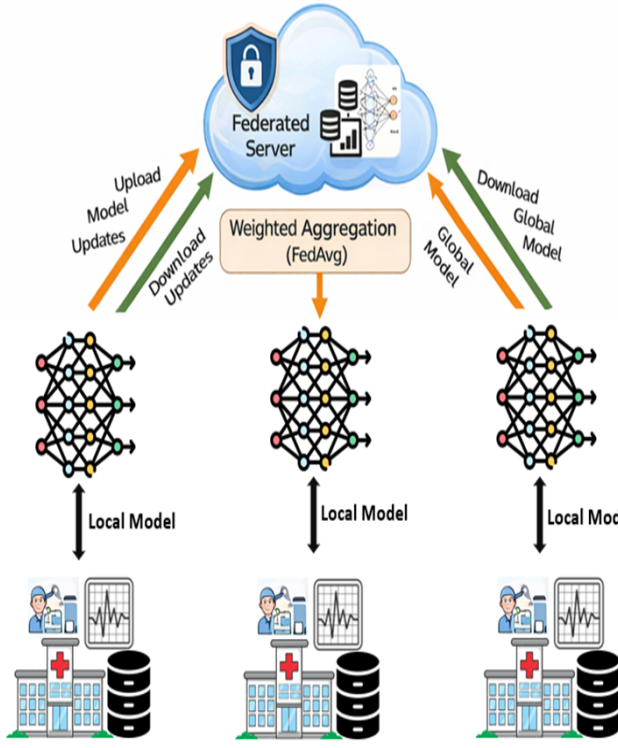


Fig. 1: Centralized federated learning framework. Clients train IEFCT-TP locally on private ECG data and upload only model weights; the server aggregates and redistributes the updated global model. Raw data never leaves the client

where $\mathcal{L}_i$ is the local cross-entropy loss and η is the learning rate.

(3) **Upload.** Client $\mathcal{C}_i$ transmits the delta $\Delta_i^{(t)} = \mathbf{w}_i^{(t+1)} - \mathbf{w}_i^{(t)}$ (or equivalently $\mathbf{w}_i^{(t+1)}$) along with its sample count n_i .

(4) **Aggregation (FedAvg).** The server forms the new global model via:

$$\mathbf{w}^{(t+1)} = \sum_{i=1}^N \frac{n_i}{n_{\text{total}}} \mathbf{w}_i^{(t+1)}, \quad n_{\text{total}} = \sum_{i=1}^N n_i. \quad (2)$$

The cycle repeats for T rounds or until a convergence criterion is satisfied.

C. Adaptive Aggregation Strategies

Beyond FedAvg, this paper also include implementation of several server-side adaptive optimizers. **FedAdam** applies momentum and per-parameter adaptive scaling to the aggregated delta $\Delta^{(t)} = \mathbf{w}_{\text{FedAvg}}^{(t+1)} - \mathbf{w}^{(t)}$:

$$\mathbf{m}^{(t)} = \beta_1 \mathbf{m}^{(t-1)} + (1 - \beta_1) \Delta^{(t)}, \quad (3)$$

$$\mathbf{v}^{(t)} = \beta_2 \mathbf{v}^{(t-1)} + (1 - \beta_2) (\Delta^{(t)})^2, \quad (4)$$

$$\mathbf{w}^{(t+1)} = \mathbf{w}^{(t)} + \eta_s \frac{\mathbf{m}^{(t)}}{\sqrt{\mathbf{v}^{(t)} + \epsilon}}. \quad (5)$$

FedAdagrad accumulates the sum of squared deltas without momentum decay, slowing the learning rate on parameters that receive frequent updates:

$$\mathbf{v}^{(t)} = \mathbf{v}^{(t-1)} + (\Delta^{(t)})^2, \quad \mathbf{w}^{(t+1)} = \mathbf{w}^{(t)} + \eta_s \frac{\Delta^{(t)}}{\sqrt{\mathbf{v}^{(t)} + \epsilon}}. \quad (6)$$

MR-MTL treats federation as multi-task learning: each client $\mathcal{C}_i$ maintains a personalized local model $\mathbf{w}_i$ and solves

$$\min_{\mathbf{w}_i} \mathcal{L}_i(\mathbf{w}_i) + \frac{\lambda}{2} \|\mathbf{w}_i - \bar{\mathbf{w}}\|_2^2, \quad \bar{\mathbf{w}} = \frac{1}{N} \sum_{i=1}^N \mathbf{w}_i, \quad (7)$$

where the regularization term λ governs the trade-off between personalization and global consistency.

IV. PROPOSED METHOD: IEFCT-TP

A. Architecture Overview

Fig. 2 depicts the full IEFCT-TP pipeline at a single client. An input ECG window $\mathbf{x} \in \mathbb{R}^L$ (with $L = 187$ samples) passes through four sequential stages: (i) dual-domain CNN feature extraction, (ii) attention-guided token pruning, (iii) transformer encoder with learnable positional encoding, and (iv) the iterative error feedback module that refines internal representations using the prediction residual. The architecture is designed to be parameter-efficient to minimize the communication overhead inherent in federated rounds.

B. Dual-Domain CNN Feature Extraction

Raw ECG beats contain diagnostic information in both the temporal and frequency domains. Rather than applying a single convolutional branch, IEFCT-TP maintains two parallel extraction paths.

Time-domain branch. A stack of three 1-D convolutional layers with filter counts [32, 64, 128], kernel size $k = 5$, batch normalization, and ReLU activation extracts local waveform features—P-wave onset and offset, QRS duration and slope, ST-segment deviation, and T-wave polarity:

$$\mathbf{H}^{\text{td}} = \text{BN}(\text{ReLU}(\mathbf{W}^{\text{td}} * \mathbf{x})) \in \mathbb{R}^{L' \times 128}, \quad (8)$$

where $*$ denotes causal 1-D convolution and $L' = \lfloor L/s \rfloor$ after max-pooling with stride $s = 2$.

Frequency-domain branch. The beat is transformed via a short-time FFT and passed through a separate [16, 32, 64] convolutional stack to capture spectral signatures—P-wave frequency content, HF/LF ratio perturbations, and QRS bandwidth—that are complementary to morphological features:

$$\mathbf{H}^{\text{fd}} = \text{BN}(\text{ReLU}(\mathbf{W}^{\text{fd}} * \mathcal{F}(\mathbf{x}))) \in \mathbb{R}^{L' \times 64}, \quad (9)$$

where $\mathcal{F}$ denotes the magnitude spectrum computed via the Fast Fourier Transform. The two feature maps are concatenated along the channel dimension and projected to a common model dimension $d_{\text{model}} = 64$ via a learned linear transformation:

$$\mathbf{Z}_0 = \mathbf{W}_{\text{proj}} [\mathbf{H}^{\text{td}} \parallel \mathbf{H}^{\text{fd}}] + \mathbf{b}_{\text{proj}}, \quad \mathbf{Z}_0 \in \mathbb{R}^{L' \times d_{\text{model}}}. \quad (10)$$

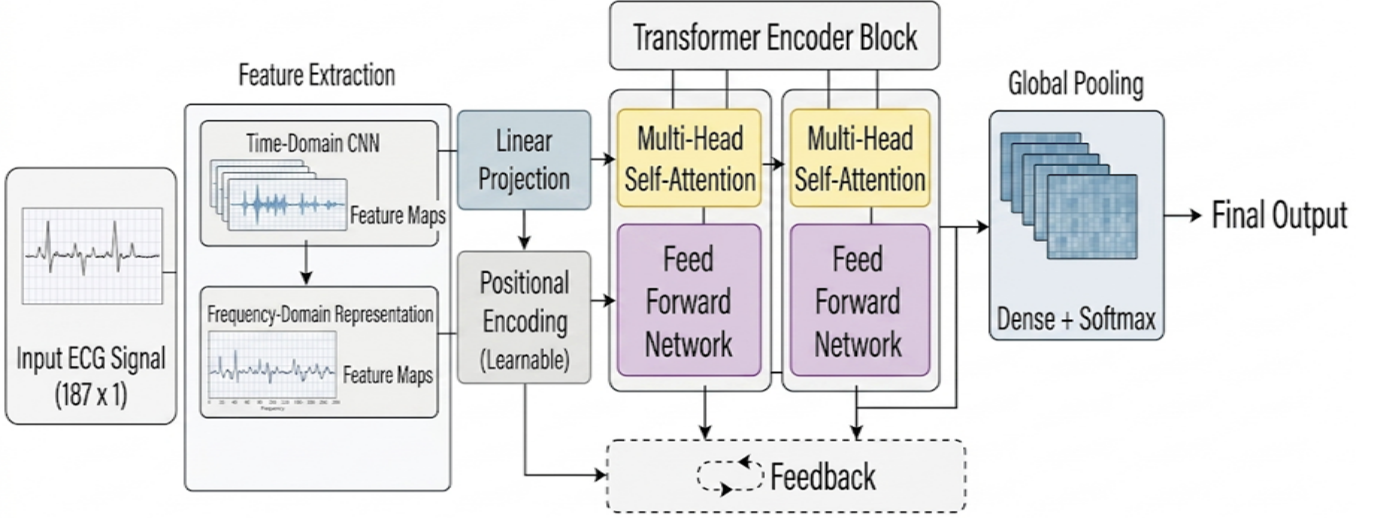


Fig. 2:

Architecture of the proposed IEFCT-TP model at the client side.

C. Attention-Score-Guided Token Pruning

Standard transformer architectures process all L' tokens uniformly regardless of their discriminative relevance. In federated settings, this wastes computation and inflates client-server communication. IEFCT-TP introduces a lightweight pruning step that retains only the K most informative tokens before the main transformer stack.

A single-head proxy attention layer computes importance scores $s = \text{softmax}(\mathbf{q}\mathbf{K}^\top/\sqrt{d})\mathbf{v}$, where $\mathbf{q}$ is a learned global query vector. The scalar salience of token l is:

$$\alpha_l = \frac{\exp(s_l)}{\sum_{l'=1}^{L'} \exp(s_{l'})}, \quad l = 1, \dots, L'. \quad (11)$$

Only the top- K tokens ranked by α_l proceed to the transformer encoders ($K = \lfloor 0.6L' \rfloor$ by default, pruning 40% of tokens):

$$\tilde{\mathbf{Z}}_0 = \text{TopK}(\mathbf{Z}_0, \{\alpha_l\}_{l=1}^{L'}) \in \mathbb{R}^{K \times d_{\text{model}}}. \quad (12)$$

This reduces the quadratic self-attention complexity from $\mathcal{O}(L'^2)$ to $\mathcal{O}(K^2)$, directly decreasing both forward-pass FLOPs and the effective model size transmitted per round.

D. Transformer Encoder with Learnable Positional Encoding

Pruned tokens are augmented with learnable positional embeddings $\mathbf{P} \in \mathbb{R}^{K \times d_{\text{model}}}$:

$$\tilde{\mathbf{Z}}_0^{\text{pos}} = \tilde{\mathbf{Z}}_0 + \mathbf{P}. \quad (13)$$

Two stacked transformer encoder layers with four attention heads then compute global sequence representations. Within each encoder block, the multi-head self-attention (MHA) sub-layer operates as:

$$\text{head}_h = \text{Attn}(\tilde{\mathbf{Z}}_0^{\text{pos}} \mathbf{W}_h^Q, \tilde{\mathbf{Z}}_0^{\text{pos}} \mathbf{W}_h^K, \tilde{\mathbf{Z}}_0^{\text{pos}} \mathbf{W}_h^V), \quad (14)$$

$$\text{MHA}(\cdot) = [\text{head}_1 \parallel \dots \parallel \text{head}_H] \mathbf{W}^O, \quad (15)$$

where $H = 4$ and each head operates in dimension $d_k = d_{\text{model}}/H = 16$. Residual connections and layer normalization follow both the MHA and feed-forward sublayers:

$$\mathbf{Z}' = \text{LayerNorm}(\tilde{\mathbf{Z}}_0^{\text{pos}} + \text{MHA}(\tilde{\mathbf{Z}}_0^{\text{pos}})), \quad (16)$$

$$\mathbf{Z}'' = \text{LayerNorm}(\mathbf{Z}' + \text{FFN}(\mathbf{Z}')). \quad (17)$$

The feed-forward network expands to $d_{\text{ff}} = 128$ dimensions before projecting back to d_{model} :

$$\text{FFN}(\mathbf{z}) = \text{ReLU}(\mathbf{z}\mathbf{W}_1 + \mathbf{b}_1) \mathbf{W}_2 + \mathbf{b}_2. \quad (18)$$

After two encoder blocks, global average pooling collapses the token dimension to yield a fixed-length representation $\mathbf{z}_{\text{pool}} \in \mathbb{R}^{d_{\text{model}}}$.

E. Iterative Error Feedback (IEF) — Core Novelty

Standard training minimizes a cross-entropy loss and propagates gradients backward through the network once per mini-batch. This works well under IID data but can stagnate in federated non-IID settings where local loss landscapes diverge substantially across clients. The Iterative Error Feedback mechanism addresses this by explicitly recycling the prediction residual as an auxiliary training signal.

Mechanism. Let $\hat{\mathbf{y}} = \text{Softmax}(\mathbf{W}_{\text{cls}} \mathbf{z}_{\text{pool}})$ be the initial class probability vector for a sample with one-hot ground-truth $\mathbf{y}$. The prediction error is:

$$\boldsymbol{\delta} = \mathbf{y} - \hat{\mathbf{y}} \in \mathbb{R}^C, \quad (19)$$

where $C = 5$ is the number of AAMI classes. This residual captures which classes are systematically over- or under-estimated by the current model state. The error is projected

into the transformer’s feature space via a learned matrix $\mathbf{W}_\delta \in \mathbb{R}^{C \times d_{\text{model}}}$:

$$\mathbf{e} = \text{ReLU}(\delta \mathbf{W}_\delta) \in \mathbb{R}^{d_{\text{model}}}. \quad (20)$$

This *feedback embedding* $\mathbf{e}$ is broadcast across the token dimension and added to the transformer’s penultimate-layer output before the pooling operation:

$$\tilde{\mathbf{Z}}'' = \mathbf{Z}'' + \mathbf{e} \cdot \mathbf{1}_K^\top. \quad (21)$$

A second forward pass then re-computes $\hat{\mathbf{y}}^{(2)} = \text{Softmax}(\mathbf{W}_{\text{cls}} \cdot \text{GAP}(\tilde{\mathbf{Z}}''))$.

Loss function. The total training objective combines both forward-pass cross-entropy terms with an optional residual magnitude regularizer:

$$\mathcal{L} = \mathcal{L}_{\text{CE}}(\hat{\mathbf{y}}, \mathbf{y}) + \gamma \mathcal{L}_{\text{CE}}(\hat{\mathbf{y}}^{(2)}, \mathbf{y}) + \mu \|\delta\|_2^2, \quad (22)$$

where $\gamma = 0.5$ weights the corrected prediction and $\mu = 0.01$ controls residual regularization. Gradients flow through both $\hat{\mathbf{y}}$ and $\hat{\mathbf{y}}^{(2)}$, so $\mathbf{W}_\delta$ learns to project diagnostically informative error components into directions of the feature space that the first-pass transformer representations underutilize.

Why IEF improves non-IID convergence. Under statistical heterogeneity, the global model initialized at each round may be poorly aligned with a given client’s data distribution, producing systematic prediction errors. Rather than relying solely on gradient updates to correct this misalignment over many local epochs, IEF provides an additional intra-batch correction that moves the representation toward the true class boundary. This is analogous to a residual correction step in iterative signal reconstruction, but here it operates in learned feature space rather than signal space. The effect is faster per-round adaptation on heterogeneous clients and reduced client drift, which is the principal source of performance degradation in non-IID federated learning.

V. EXPERIMENTAL SETUP

A. Datasets

MIT-BIH Arrhythmia Database [11]: 48 half-hour ambulatory two-lead ECG recordings from 47 patients sampled at 360 Hz. Annotations are mapped to the AAMI five-class schema: N (normal sinus), S (supraventricular ectopic), V (ventricular ectopic), F (fusion), and Q (unclassifiable). Individual beats are segmented as 187-sample windows centered on annotated R-peaks (90 samples before and 97 after).

St. Petersburg INCART Database: 75 annotated 12-lead, 30-minute ECG records sampled at 257 Hz. The proposed method extract the lead II channel and apply equivalent windowing and AAMI mapping to assess cross-database generalization.

Preprocessing steps common to both datasets include: (i) bandpass filtering (0.5–40 Hz) to suppress baseline wander and high-frequency muscle noise; (ii) per-window min-max normalization to $[0, 1]$; (iii) random over-sampling of minority classes (V, F) to address severe class imbalance in MIT-BIH.

B. Federated Data Partitions

IID: Beats are randomly shuffled and equally distributed among clients. Each client’s class distribution approximates the population distribution.

Non-IID: Records are assigned to clients by patient identifier, producing realistic cross-institutional heterogeneity: certain clients may lack V or F class examples entirely, simulating a hospital serving a predominantly healthy population.

Experiments are conducted with $N \in \{4, 5, 6\}$ clients and $T = 5$ federated rounds per experiment. Each local training phase runs for $E = 5$ epochs with batch size 64 and learning rate $\eta = 10^{-3}$.

C. Baselines

The paper compare IEFCT-TP against five established architectures, each implemented under identical federated conditions:

- **RNN:** Vanilla recurrent network, two layers of 64 units, susceptible to vanishing gradients.
- **LSTM:** Bidirectional LSTM (64 units) $\rightarrow$ LSTM (32 units) $\rightarrow$ Dense(64) $\rightarrow$ Softmax; captures forward and backward dependencies.
- **CNN:** Two Conv1D layers (64 filters, kernels 3 and 1) $\rightarrow$ Flatten $\rightarrow$ Dense(118) $\rightarrow$ Softmax; efficient baseline.
- **ResNet-1D:** ResNet-34-inspired network with four residual blocks (64 $\rightarrow$ 128 $\rightarrow$ 256 $\rightarrow$ 512 filters), global average pooling, Dense(256), Dropout(0.3), Softmax.
- **Transformer:** Two stacked transformer encoder blocks with 4-head MHA, linear projection to 64-dim, learnable positional encoding, global average pooling, Dense(64), Dropout(0.3), Softmax—identical to the transformer component of IEFCT-TP without CNN branches, token pruning, or IEF.

D. Evaluation Metrics

Model performance is quantified by accuracy (Acc), macro-averaged precision (P), recall (R), and F1-score (F1), computed as:

$$\begin{aligned} \text{Acc} &= \frac{TP + TN}{TP + TN + FP + FN}, & P &= \frac{TP}{TP + FP}, \\ R &= \frac{TP}{TP + FN}, & F1 &= \frac{2PR}{P + R}. \end{aligned} \quad (23)$$

VI. RESULTS AND DISCUSSION

A. Comparison of Deep Learning Architectures

Table I reports performance under 5-client IID federation with FedAvg. RNN and LSTM underperform expectably: sequential processing limits long-range dependency capture and both are prone to gradient vanishing in federated rounds where gradients accumulate over multiple local epochs. ResNet-1D benefits from residual connections but lacks the attention mechanism needed to weight clinically relevant beats. CNN achieves 96% accuracy through efficient local morphology capture. The Transformer outperforms CNN with 98%, indicating that parallel global attention on the whole beat window generates more informative features compared with the solely

convolutional methods. The proposed IEFCT-TP method outperforms all other baseline approaches with **99.3% accuracy and 98.95% F1 score**, benefitting from the joint contribution of the two domains, token pruning, and IEF.

TABLE I: Performance Comparison of Deep Learning Architectures (FedAvg, 5 Clients, IID)

Model	Acc (%)	P (%)	R (%)	F1 (%)
RNN	75	68	75	71
LSTM	76	68	76	72
ResNet-1D	83	85	83	84
CNN	96	95	92	94
Transformer	98	94	95	94
IEFCT-TP (Proposed)	99.3	98.6	99.3	98.95

B. Performance Under IID vs. Non-IID Settings

Tables II, III, and IV provide results on accuracy and F1 under both data settings for CNN, Transformer, and IEFCT-TP respectively with different client numbers $N = \{4, 5, 6\}$.

With the IID setting, all three models exhibit reliable high performance, with IEFCT-TP always at $\approx 99\%$. The cost of increasing the number of clients is rather small since each new client brings a representation sample of the whole distribution, which ensures the variance of the FedAvg averaging is still very low.

In contrast, the Non-IID results show a noticeable distinction between the methods. The CNN model performs poorly, dropping to 89–90% due to purely local feature-extraction being unable to cover the problem of class absence in some client data. However, due to global attention, the Transformer model demonstrates relatively decent performance: 89% accuracy with 69% recall in case of 6 clients. IEFCT-TP shows even better results with up to **95.7% accuracy** in case of 5 clients with Non-IID split. The reason for that is the use of the IEF module, which provides intra-batch correction depending on the current error.

TABLE II: IID vs. Non-IID Performance: CNN (FedAvg)

Data	Clients	Acc	P	R	F1
IID	4	94	95	96	95
	5	96	95	92	94
	6	94	94	95	95
Non-IID	4	89	77	62	68
	5	90	91	73	82
	6	90	84	63	73

C. Comparison of Federated Optimization Algorithms

Table V reports the four federated optimization algorithms applied to IEFCT-TP under 5-client IID partitions. All strategies achieve high accuracy with IEFCT-TP, confirming the architecture’s robustness to the choice of aggregation rule. FedAvg achieves the best overall balance (99.3% Acc, 98.95% F1). FedAdam converges with slightly higher iteration-to-iteration variance, occasionally overshooting, but recovers to 99.0% Acc. MR-MTL provides the second-highest F1 at

TABLE III: IID vs. Non-IID Performance: Transformer (FedAvg)

Data	Clients	Acc	P	R	F1
IID	4	96	95	96	95
	5	98	94	95	94
	6	95	95	95	95
Non-IID	4	92	90	91	90
	5	88	87	86	87
	6	89	84	78	81

TABLE IV: IID vs. Non-IID Performance: IEFCT-TP (FedAvg)

Data	Clients	Acc	P	R	F1
IID	4	98.67	97.35	98.67	98.00
	5	99.30	98.60	99.30	98.95
	6	98.17	96.37	98.17	97.26
Non-IID	4	94.00	88.00	82.00	85.00
	5	95.70	90.80	85.40	87.00
	6	90.00	91.00	90.00	90.00

98.75% by exploiting the regularization toward the global mean to prevent individual clients from straying into degenerate local optima—an advantage that becomes more pronounced under non-IID splits. FedAdagrad, while slightly lower in accuracy (98.5%), remains a robust choice when client learning rates need per-parameter adaptation.

TABLE V: Federated Optimization Algorithm Comparison: IEFCT-TP (5 Clients, IID)

Algorithm	Acc (%)	P (%)	R (%)	F1 (%)
FedAvg	99.30	98.60	99.30	98.95
FedAdam	99.00	98.01	99.00	98.50
FedAdagrad	98.50	97.02	98.50	97.76
MR-MTL	99.17	98.34	99.17	98.75

D. Cross-Dataset Generalization

Table VI shows that IEFCT-TP generalizes well to the St. Petersburg database, achieving 99.27% accuracy—marginally below the MIT-BIH result, largely attributable to differences in sampling rate (257 vs. 360 Hz) and lead configuration. All models exhibit a small but consistent performance drop on St. Petersburg, validating the difficulty of cross-database ECG classification and confirming that the relative performance order holds across datasets.

E. Comparison with Literature

The work by IEFCT-TP is compared to other recently published works in Table VII. Although Mane *et al.* [8] claimed an accuracy of 99.0%, their recall rate of 91.0% signifies that the test set contains imbalanced classes. Mehta and Rathour [7] secured a high precision value (97.8%) at the expense of a low recall rate (88.0%), indicating that their algorithm is prone to overpredicting the dominant normal class. IEFCT-TP secured the highest recall rate of **99.3%**.

TABLE VI: Cross-Dataset Performance (5 Clients, FedAvg, IID)

Dataset	Model	Acc	P	R	F1
MIT-BIH	IEFCT-TP	99.3	98.6	99.3	98.95
	Transformer	98.0	94.0	95.0	94.0
	CNN	96.0	95.0	92.0	94.0
	ResNet-1D	83.0	85.0	83.0	84.0
St. Petersburg	IEFCT-TP	99.27	99.29	99.25	99.27
	Transformer	96.0	95.0	96.0	96.0
	CNN	95.0	94.0	95.0	94.0
	ResNet-1D	85.0	83.0	84.0	84.0

TABLE VII: Comparison with Prior Federated ECG Classification Studies

Study	Acc	P	R	F1
Mehta & Rathour [7]	98.5	97.8	88.0	92.6
Çelik & Güllü [5]	95.0	94.2	89.5	91.8
Mane <i>et al.</i> [8]	99.0	97.0	91.0	93.9
ElBedoui [9]	93.0	92.5	88.7	90.6
Weimann & Conrad [6]	97.0	96.2	90.1	92.9
IEFCT-TP (Proposed)	99.3	98.6	99.3	98.95

F. Accuracy and Loss Convergence

Fig. 3 illustrates the training and test accuracy/loss trends through five federated epochs using the IEFCT-TP approach under the traditional IID configuration with five clients. Both trends increase sharply from round 1 to round 3 due to fast local adaptation made possible by the IEF process; the trends level off close to 99% by round 4. The close correspondence of training and testing curves indicates that there is no client overfitting. The associated loss trends decrease steadily on both datasets to reach a test loss of ≈ 0.05 by round 5.

VII. CONCLUSION

The proposed framework, IEFCT-TP, is a federated deep learning system capable of preserving privacy in ECG classifications. There are three key design considerations that contribute to its superior performance. The first is the presence of a two-stream CNN frontend (time-domain and spectral), which allows it to learn complementary local feature representations that are not learnable from the same stream. The second is the use of an attention score-based token pruning technique, where 40% of the tokens are pruned before the transformer layer. This allows for a reduction in the quadratic nature of the attention computation and the size of the model updates sent over the federation during each round. The third and most important one is the Iterative Error Feedback, which feeds the prediction residuals back into the learning process.

In experiments using two electrocardiogram datasets, six baseline architectures, four aggregation methods, and two classes of datasets—both independent and identically distributed (IID) and non-independent and identically distributed (non-IID)—IEFCT-TP outperformed all other approaches, recording an accuracy and F1 score of up to 99.3% and 98.95%, respectively, under FedAvg using five clients. It must be noted that IEFCT-TP is the only method that managed to

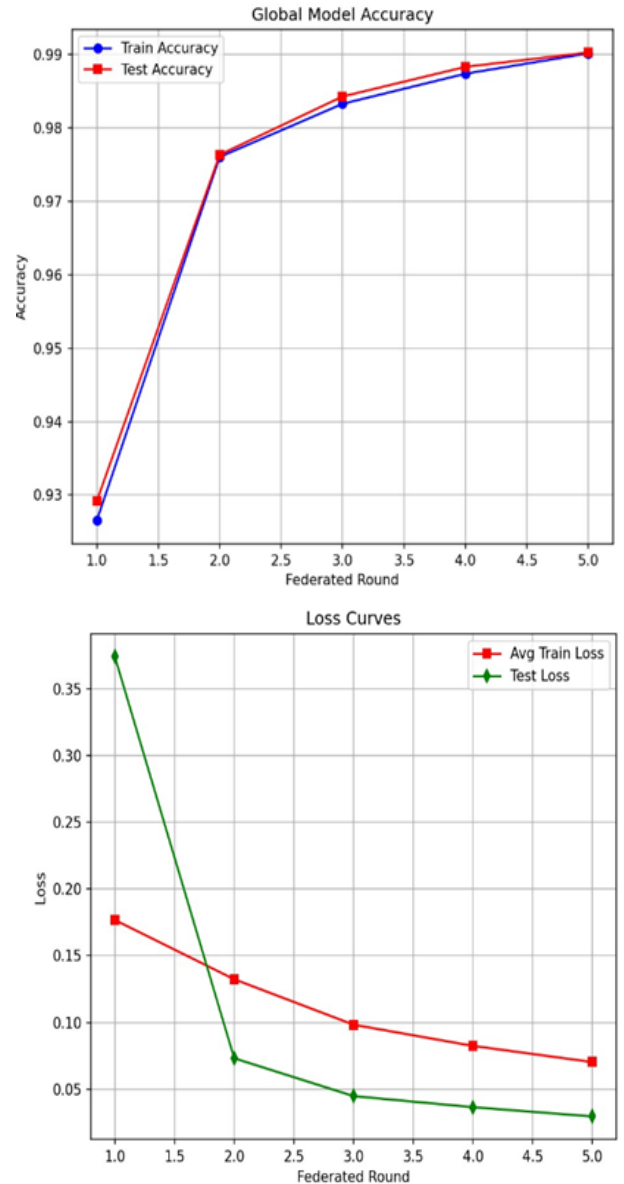


Fig. 3: IEFCT-TP training and testing accuracy/loss curves for five federated iterations using 5-client IID partition and FedAvg algorithm. Perfect train/test correlation indicates no overfitting and successful federated training.

score above 99% in both accuracy and recall, which is of great importance when considering applications in the clinical screening context, as lives could hang in the balance in such cases.

VIII. FUTURE WORK

There are a few possible future directions to pursue here. A full decentralization or blockchain-based coordination scheme would solve the problem of single point of failure and provide further assurances. The use of differential privacy at the gradient stage, by adding calibrated Gaussian noise before

uploading, and homomorphic encryption for secure aggregation would mitigate the problem of inference attacks related to the sharing of gradient information. Expanding the framework to include a 12-lead ECG input via the lead-wise attention fusion technique [15] would allow federated training on more complex input patterns commonly encountered in hospitals. From the optimization perspective, the use of MAML-based initialization of the models could reduce the amount of epochs needed at the client stage to achieve convergence. Last but not least, hardware validation on edge devices such as Raspberry Pi and FPGA and deployment within hospital networks with the IRB-agreed data sharing agreement is needed.

REFERENCES

- [1] World Health Organization, "Cardiovascular diseases (CVDs)," *WHO Fact Sheet*, 2021.
- [2] P. Rajpurkar *et al.*, "Cardiologist-level arrhythmia detection and classification in ambulatory electrocardiograms using a deep neural network," *Nature Medicine*, vol. 25, pp. 65–69, 2019.
- [3] R. Miotto, F. Wang, S. Wang, X. Jiang, and J. T. Dudley, "Deep learning for healthcare: review, opportunities and challenges," *Briefings in Bioinformatics*, vol. 19, no. 6, pp. 1236–1246, 2018.
- [4] H. B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. A. Y. Arcas, "Communication-efficient learning of deep networks from decentralized data," in *Proc. AISTATS*, 2017, pp. 1273–1282.
- [5] E. Çelik and M. K. Güllü, "Comparison of federated learning strategies on ECG classification," in *Proc. 2023 Innovations in Intelligent Systems and Applications Conference (ASYU)*, Sivas, Turkey, 2023, pp. 1–4, doi: 10.1109/ASYU58738.2023.10296796.
- [6] K. Weimann and T. O. F. Conrad, "Federated learning with deep neural networks: a privacy-preserving approach to enhanced ECG classification," *IEEE J. Biomed. Health Inform.*, vol. 28, no. 11, pp. 6931–6943, Nov. 2024, doi: 10.1109/JBHI.2024.3427787.
- [7] S. Mehta and A. Rathour, "ECG data security and privacy: leveraging federated learning for better classification," in *Proc. 2024 4th Int. Conf. Intelligent Technologies (CONIT)*, Bangalore, India, 2024, pp. 1–6, doi: 10.1109/CONIT61985.2024.10627749.
- [8] D. Mane, J. Jain, U. Jaju, K. Agarwal, and A. Kalamkar, "A federated approach towards detecting ECG arrhythmia," in *Proc. 2024 2nd Int. Conf. Sustainable Computing and Smart Systems (ICSCSS)*, Coimbatore, India, 2024, pp. 106–111.
- [9] K. ElBedoui, "ECG classification based on federated unlearning," in *Proc. 2023 Int. Symp. Networks, Computers and Communications (ISNCC)*, Doha, Qatar, 2023, pp. 1–5, doi: 10.1109/ISNCC58260.2023.10323758.
- [10] N. Asif, S. Dewangan, A. Dhaliwal, and A. Shukla, "Detecting electrocardiogram arrhythmia empowered with weighted federated learning," *IEEE Access*, vol. 12, pp. 55123–55134, 2024.
- [11] G. B. Moody and R. G. Mark, "The impact of the MIT-BIH arrhythmia database," *IEEE Eng. Med. Biol. Mag.*, vol. 20, no. 3, pp. 45–50, 2001.
- [12] L. Zhou, J. Liu, and S. Li, "Hierarchical federated learning with social context clustering-based participant selection for IoMT applications," *IEEE Trans. Comput. Social Syst.*, vol. 10, no. 4, pp. 1458–1469, Aug. 2023.
- [13] S. Zhang, C. Lian, B. Xu, J. Zang, and Z. Zeng, "A token selection-based multi-scale dual-branch CNN-transformer network for 12-lead ECG signal classification," *Knowledge-Based Syst.*, vol. 280, 2023, Art. no. 111006.
- [14] M. Niloy *et al.*, "Lead-aware multi-resolution transformer with domain adaptation for beat-level ECG arrhythmia classification," *IEEE Open J. Comput. Soc.*, vol. 6, pp. 1946–1948, 2025.
- [15] R. V. Sharan, H. Xiong, and H. Cui, "Cardiac abnormality detection from 12-lead ECG using multibranch CNN and attention-based feature fusion," *IEEE Trans. Instrum. Meas.*, vol. 74, 2025, Art. no. 4019210.
- [16] S. Chon, K.-W. Ha, S. Park, and S. Jung, "An ECG beat classification method using multi-kernel ResNet with transformer," in *Proc. IEEE Int. Conf. Big Data and Smart Computing (BigComp)*, 2023, pp. 140–141.
- [17] K. Liu, R. Yang, L. Qi, W. Luan, and Z. Zhang, "ART-Net: an attention-based hybrid ResNet-Transformer network for 12-lead ECG signal classification," in *Proc. IEEE Int. Conf. Systems, Man, and Cybernetics (SMC)*, 2024, pp. 3664–3665.
- [18] H. I. Bulbul, N. Usta, and M. Yildiz, "Classification of ECG arrhythmia with machine learning techniques," in *Proc. 16th IEEE Int. Conf. Machine Learning and Applications (ICMLA)*, Cancun, Mexico, 2017, pp. 546–549.
- [19] T. Subba and T. Chingtham, "Comparative analysis of machine learning algorithms with advanced feature extraction for ECG signal classification," *IEEE Access*, vol. 12, pp. 57727–57740, 2024.
- [20] C. Dwork and A. Roth, "The algorithmic foundations of differential privacy," *Foundations and Trends in Theoretical Computer Science*, vol. 9, no. 3–4, pp. 211–407, 2014.
- [21] A. Fallah, A. Mokhtari, and A. Ozdaglar, "Personalized federated learning with theoretical guarantees: a model-agnostic meta-learning approach," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2020, vol. 33, pp. 3557–3568.